CHAPTER 7

INTEGRALS

CHAPTER ESSENCE AT A GLANCE:
Indefinite and definite integrals, integration by substitution, parts, and partial fractions, and the fundamental theorem of calculus.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Fundamental Theorem of Calculus:** Definite integration computes net accumulated area and serves as the inverse operation to differentiation, linking derivatives to antiderivative differences across interval endpoints.

• **Method of Substitution:** Transforms complex composite integrands into standard elementary integrals by introducing an auxiliary variable whose differential matches an existing factor in the integrand.

• **Integration by Parts Priority:** Evaluates products of two functions using the standard inverse trigonometric, logarithmic, algebraic, trigonometric, and exponential priority rule to select the factor for differentiation.

• **Partial Fractions Decomposition:** Resolves proper rational algebraic quotients into sums of simpler partial fractions having linear or irreducible quadratic denominators, enabling term-by-term integration.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$\int u \cdot v \, dx = u \int v \, dx - \int \left(\frac{du}{dx} \int v \, dx \right) dx$$

Integration by parts formula using standard ILATE priority

$$\int e^x [f(x) + f'(x)] \, dx = e^x f(x) + C$$

Special exponential product integration theorem

$$\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right| + C, \quad \int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + C$$

Standard quadratic rational antiderivatives

$$\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \left(\frac{x}{a} \right) + C, \quad \int \frac{dx}{\sqrt{x^2 \pm a^2}} = \ln \left| x + \sqrt{x^2 \pm a^2} \right| + C$$

Standard square root quadratic integrals

$$\int_a^b f(x) \, dx = \int_a^b f(a+b-x) \, dx \rightarrow \int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} \, dx = \frac{\pi}{4}$$

King Property of definite integrals with standard evaluation

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** [Trap Alert: The 2I Factor] In applying King's property P4, students frequently forget that adding I + I gives 2I! Dividing by 2 at the end is mandatory to avoid losing 1 Mark.

• **Alert:** [Trap Alert: Partial Fraction Degree] Always ensure numerator degree < denominator degree before applying partial fractions. If degree(num) ≥ degree(den), perform polynomial long division first!

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Evaluate by substitution: $\int \frac{(2x)}{(1+x^2)}$ dx and $\int ((\log x)^2) / (x)$ dx. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Substitution Method]** 1. $I_1 = \int$ [2x over $(1 + x^2)$] dx. Put $t = 1 + x^2 \Rightarrow dt = 2x\,dx$. $I_1 = \int dt$ over $t = \ln|t| + C = \ln(1 + x^2) + C$. [1 Mark]

• 2. $I_2 = \int$ [$(\ln x)^2$ over x] dx. Put $t = \ln x$ implies $dt = (1 \text{ over } x)\,dx$. $I_2 = \int t^2\,dt = t^3$ over $3 + C = (1 \text{ over } 3)(\ln x)^3 + C$. [1 Mark]

• **[Boxed Result: $I_1 = \ln(1 + x^2) + C$ and $I_2 = (1 \text{ over } 3)(\ln x)^3 + C$]**

2. Evaluate: $\int \sin(2x + 5)\,dx$ and $\int (\cos(\sqrt{x})) / (\sqrt{x})\,dx$. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Evaluations]** 1. $\int \sin(2x + 5)\,dx = -\cos(2x + 5)$ over $2 + C$. [1 Mark]

• 2. $\int$ [cos(sqrt(x)) over sqrt(x)] dx: Put $t = \sqrt{x}$ implies $dt = 1$ over $(2\sqrt{x})\,dx$ implies $(1 \text{ over } \sqrt{x})\,dx = 2\,dt$. $\int \cos t\,(2\,dt) = 2 \sin t + C = 2 \sin(\sqrt{x}) + C$. [1 Mark]

• **[Boxed Result: $-(1 \text{ over } 2)\cos(2x + 5) + C$ and $2 \sin(\sqrt{x}) + C$]**

3. Evaluate: (a) $\int (2x^2 + e^x)\,dx$, (b) $\int \sec x (\sec x + \tan x)\,dx$, (c) $\int \frac{(1 - \sin x)}{(\cos^2 x)}$ dx. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Standard Integrals Evaluation]** (a) $\int (2x^2 + e^x)\,dx = 2\,(x^3 \text{ over } 3) + e^x + C = (2 \text{ over } 3)\,x^3 + e^x + C$. [1 Mark]

• (b) $\int \sec x (\sec x + \tan x)\,dx = \int (\sec^2 x + \sec x \tan x)\,dx = \tan x + \sec x + C$. [1 Mark]

• (c) $\int$ [$(1 - \sin x)$ over $\cos^2 x$] $dx = \int (\sec^2 x - \sec x \tan x)\,dx = \tan x - \sec x + C$. [1 Mark]

• **[Boxed Result: (a) $(2 \text{ over } 3)x^3 + e^x + C$ | (b) $\tan x + \sec x + C$ | (c) $\tan x - \sec x + C$]**

4. Evaluate: $\int \sin^2 x\,dx$ and $\int \cos^3 x\,dx$. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Trigonometric Integrals]** 1. $\int \sin^2 x\,dx = \int$ [$(1 - \cos 2x)$ over 2] $dx = (1 \text{ over } 2)$ [$x - \sin 2x$ over 2] $+ C = x$ over $2 - \sin 2x$ over $4 + C$. [1.5 Marks]

• 2. $\int \cos^3 x\,dx = \int \cos^2 x \cdot \cos x\,dx = \int (1 - \sin^2 x)\cos x\,dx$. Put $t = \sin x$ implies $dt = \cos x\,dx$: $\int (1 - t^2)\,dt = t - t^3$ over $3 + C = \sin x - (1 \text{ over } 3)\sin^3 x + C$. [1.5 Marks]

• **[Boxed Result: (a) x over $2 - (\sin 2x)$ over $4 + C$ | (b) $\sin x - (1 \text{ over } 3)\sin^3 x + C$]**

5. Evaluate: $\int \sin 2x \cos 3x\,dx$. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Product-to-Sum Transformation]** $2 \sin A \cos B = \sin(A + B) + \sin(A - B)$. $\sin 2x \cos 3x = (1 \text{ over } 2)$ [$2 \cos 3x \sin 2x$] = (1 over 2) [$\sin(3x + 2x) - \sin(3x - 2x)$] = (1 over 2) [$\sin 5x - \sin x$]. [1.5 Marks]

• **[Integration]** $\int \sin 2x \cos 3x\,dx = (1 \text{ over } 2) \int (\sin 5x - \sin x)\,dx = (1 \text{ over } 2)$ [$-\cos 5x$ over $5 - (-\cos x)$] $+ C = -\cos 5x$ over $10 + \cos x$ over $2 + C$. [1.5 Marks]

• **[Boxed Result: $(1 \text{ over } 2)\cos x - (1 \text{ over } 10)\cos 5x + C$]**

6. Integrate by partial fractions: $\int (2x) / ((x^2 + 1)(x^2 + 2))\,dx$. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Substitution & Decomposition]** Put $t = x^2 \Rightarrow dt = 2x\,dx$. $\int$ [dt over $((t + 1)(t + 2))$]. 1 over [$(t + 1)(t + 2)$] = 1 over $(t + 1) - 1$ over $(t + 2)$. [2 Marks]

• **[Integration]** $\int$ [1 over $(t + 1) - 1$ over $(t + 2)$] $dt = \ln|t + 1| - \ln|t + 2| + C = \ln|(t + 1) \text{ over } (t + 2)| + C$. Substituting back $t = x^2$: $= \ln|(x^2 + 1) \text{ over } (x^2 + 2)| + C$. [2 Marks]

• **[Boxed Result: $\ln|(x^2 + 1) \text{ over } (x^2 + 2)| + C$]**

7. Evaluate: ∫ sqrt($a^2 - x^2$) dx. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Derivation of ∫ √($a^2 - x^2$) dx]** Put x = a sin θ ⇒ dx = a cos θ dθ. √($a^2 - x^2$) = a cos θ. ∫ √($a^2 - x^2$) dx = ∫ (a cos θ)(a cos θ dθ) = a^2 ∫ $\cos^2$ θ dθ = a^2 ∫ [(1 + cos 2θ) over 2] dθ = (a^2 over 2) [θ + sin 2θ over 2] = (a^2 over 2) [θ + sin θ cos θ] + C. [2.5 Marks]

• Since sin θ = x over a, cos θ = sqrt($a^2 - x^2$) over a, and θ = $\sin^{-1}$(x over a): = (a^2 over 2) [$\sin^{-1}$(x over a) + (x over a) (sqrt($a^2 - x^2$) over a)] + C = (x over 2) sqrt($a^2 - x^2$) + (a^2 over 2) $\sin^{-1}$(x over a) + C. [1.5 Marks]

• **[Boxed Result: ∫ √($a^2 - x^2$) dx = (x over 2) √($a^2 - x^2$) + (a^2 over 2) $\sin^{-1}$(x over a) + C]**